



MANIPAL
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COURSE NAME	INTRODUCTION TO WEB PROGRAMMING

Set-I

Answer 1 :

Main Features, Uses, and Versions of HTML:

- **Features:** HTML (HyperText Markup Language) provides a framework for structuring content on the web, supporting multimedia, links, and styling.
- **Uses:** Designing web pages, embedding multimedia, and linking multiple pages.
- **Versions:** From HTML 1.0 to HTML5, with HTML5 supporting advanced features like semantic tags, multimedia elements, and offline storage.

Syntax Structure:

HTML documents consist of:

1. **Head Section:** Contains metadata, title, and links to external resources.

html

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<head>

<title>Sample Page</title>

</head>

2. **Body Section:** Includes content like headings, paragraphs, images, and more.

html

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```
<body>

  <h1>Welcome to HTML</h1>

  <p>This is a paragraph.</p>

</body>
```

Answer 2 :

Creating Tables in HTML:

Tables are created using the <table> tag with rows (<tr>) and data cells (<td>).

Example:

html

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```
<table border="1">

  <tr>

    <th>Name</th>

    <th>Age</th>

  </tr>

  <tr>

    <td>John</td>

    <td>25</td>

  </tr>

</table>
```

Purpose of List and Text Styling Tags:

- **List Tags:** (unordered list), (ordered list), (list item).
- **Text Styling Tags:** (bold), <i> (italic), <u> (underline), <mark> (highlight).

Answer 3 :

CSS and Applying Simple Styles:

- **Definition:** CSS (Cascading Style Sheets) is used to style HTML elements, separating design from content.

Example of Applying Styles:

html

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```
<style>
```

```
h1 {  
    color: blue;  
    font-size: 20px;  
}  
</style>
```

Inline Styles:

```
html  
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<h1 style="color: red;">Styled Heading</h1>
```

Set-II

Answer 4 :

Inserting and Aligning Text in HTML:

Text can be inserted using tags like <p>, <h1> to <h6>, and alignment can be controlled using the align attribute or CSS.

Example:

```
html  
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<p align="center">This is centered text.</p>  
<h1 style="text-align: left;">Left-aligned heading</h1>
```

Incorporating Images and Hyperlinks:

- **Images:** Use the tag.
Example:

```
html  
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```

- **Hyperlinks:** Use the <a> tag.
Example:

```
html  
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<a href="https://example.com">Visit Example</a>
```

Answer 5 :

Structure of HTML Forms:

Forms collect user input through elements like text boxes, buttons, and dropdowns.

Components:

1. **Form Elements:** <form> defines the form, with attributes like action and method.
2. **Input Types:** <input> supports text, password, checkbox, radio buttons, etc.
3. **Form Attributes:** Include placeholder, name, and required.

Example:

html

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```
<form action="/submit" method="post">
  <label for="name">Name:</label>
  <input type="text" id="name" name="name" required>
  <input type="submit" value="Submit">
</form>
```

Enhancing User Interaction:

- **Validation:** Attributes like required ensure proper data entry.
- **Labels:** Improve accessibility by associating labels with inputs.
- **Interactive Features:** Dropdowns and buttons simplify user tasks.

Answer 6 :

Basic Syntax Rules, Data Types, and Operators in JavaScript:

1. **Syntax Rules:**
 - Statements end with ;.
 - Case-sensitive.
 - Variables declared with var, let, or const.
2. **Data Types:**
 - **Primitive:** String, Number, Boolean, Null, Undefined.
 - **Non-Primitive:** Object, Array.
3. **Operators:**
 - **Arithmetic:** +, -, *, /.
 - **Comparison:** ==, ===, !=, >=.
 - **Logical:** &&, ||, !.

JSON in JavaScript:

- **Definition:** JSON (JavaScript Object Notation) is a lightweight data format for storage and exchange.
- **Parsing and Stringifying:**

javascript

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```
let jsonString = '{"name": "John", "age": 25}';
```

```
let obj = JSON.parse(jsonString); // Parse JSON to object
```

```
let newJson = JSON.stringify(obj); // Convert object to JSON string
```